

MICHAEL CHADOLIAS

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Physics graduate with experience in building machine learning pipelines, data preprocessing, and model optimisation. Skilled in Python, statistical analysis, and scalable computational workflows for large datasets.

EXPERIENCE

Erlangen Centre for Astroparticle Physics (ECAP)

Master Thesis Student

Erlangen, Germany

Aug 2023 – Oct 2024, Full-time

- Engineered a scalable, automated data pipeline for preprocessing 500k+ simulation files, improving speed throughput by 20% and reducing storage by 35%.
- Developed automated, parallel workflows (bash + SLURM) for large-scale parameter sweeps on HPC clusters (NHR@FAU, CC@Lyon), enabling reproducible simulation and analysis runs.
- Executed large-scale parameter sweeps across 200+ model configurations using reproducible workflows, evaluating model performance and systematic uncertainties.

Erlangen Centre for Astroparticle Physics (ECAP)

Graduate Research Assistant

Erlangen, Germany

Nov 2022 – Apr 2023, Part-time

- Engineered a robust Monte Carlo production pipeline using Snakemake, resulting in a reduction of processing errors by 30% and establishing a more reliable framework for complex data simulations in research projects.
- Diagnosed and optimised distributed workflow executions, reducing job failures and improving reproducibility on HPC environments.
- Optimised SLURM resource profiles, achieving a 25% CPU workload reduction through improved parallelisation and job scheduling.

PROJECTS

Detect Pulsar API | Python, FastAPI, Docker, XGBoost, SHAP

🌐 mchadolias/detect-pulsar-api

- Built an end-to-end supervised learning pipeline for blazar source classification, including data preprocessing, feature engineering, and model benchmarking with XGBoost.
- Developed a production-ready FastAPI prediction service and containerised the entire application with Docker for deployment.
- Applied SHAP-based explainability to interpret model behaviour, evaluate feature contributions, and ensure model robustness.

MiniBooNE Event Classifier | Python, scikit-learn, Optuna, MLflow

🌐 mchadolias/miniboone-classifier

- Applied supervised learning methods for binary classification of particle events, including preprocessing, feature scaling, and algorithm comparison.
- Performed hyperparameter optimisation using ASHA via Optuna to efficiently explore the model search space and improve predictive performance.
- Tracked experiments and model artefacts with MLflow to enable reproducible workflows and systematic evaluation.

SKILLS

Programming: Python, C/C++, Bash, SQL, NumPy, Pandas, Matplotlib, Seaborn, Plotly

ML & Data Tools: scikit-learn, XGBoost, Tensorflow, PyTorch, SHAP, Optuna, MLflow, Snakemake

Deployment & DevOps: FastAPI, Flask, Docker, Apptainer, CI/CD (GitHub Actions/Gitlab), SLURM

Foreign Languages: Greek (Native), English (C2), German (B2), Italian (A1)

EDUCATION

Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)

M.Sc. in Physics, Astroparticle Specialisation — Grade: 90/100

Erlangen, Germany

Apr 2022 – Feb 2025

Aristotle University of Thessaloniki (AUTH)

B.Sc. in Physics — Grade: 7.84 / 10.00

Thessaloniki, Greece

Oct 2017 – Feb 2022

CERTIFICATIONS

- **ML Zoomcamp (DataTalks.Club)** — In Progress — Expected Jan 2026